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Spatial pattern formation in a chemotaxis–diffusion–growth model

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ABSTRACT

Mimura and one of the authors (1996) proposed a mathematical model for the pattern dynamics of aggregating regions of biological individuals possessing the property of chemotaxis. For this model, Tello and Winkler (2007) [22] obtained infinitely many local branches of nonconstant stationary solutions bifurcating from a positive constant solution, while Kurata et al. (2008) numerically showed several spatio-temporal patterns in a rectangle. Motivated by their work, we consider some qualitative behaviors of stationary solutions from global and local (bifurcation) viewpoints in the present paper. First we study the asymptotic behavior of stationary solutions as the chemotactic intensity grows to infinity. Next we construct local bifurcation branches of stripe and hexagonal stationary solutions in the special case when the habitat domain is a rectangle. For this case, the directions of the branches near the bifurcation points are also obtained. Finally, we exhibit several numerical results for the stationary and oscillating patterns.

1. Introduction

In this paper we are concerned with the following reaction–diffusion–advection system:

$$
\text{(P)}\quad
\begin{cases}
\dfrac{\partial u}{\partial t} = d\Delta u - \alpha\nabla\cdot(u\nabla\rho) + f(u) & \text{in } \Omega\times(0,\infty),\\[2mm]
\dfrac{\partial \rho}{\partial t} = \Delta\rho - b\rho + cu & \text{in } \Omega\times(0,\infty),\\[2mm]
\dfrac{\partial u}{\partial \nu} = \dfrac{\partial \rho}{\partial \nu} = 0 & \text{on } \partial\Omega\times(0,\infty),\\
u(x,0) = u_0(x), \qquad \rho(x,0) = \rho_0(x) & \text{in } \Omega.
\end{cases}
$$

Here Ω is a bounded domain in $\boldsymbol{R}^N$ with smooth boundary $\partial\Omega$; b, c, d and α are positive constants. The system (P) is a mathematical model, which was proposed by Mimura and one of the authors [1], for the pattern dynamics of aggregating regions of biological individuals possessing the property of chemotaxis. Then the unknown functions $u(x,t)$ and $\rho(x,t)$ denote the population density of biological individuals and the concentration of a chemical substance, which is produced by the individual itself, at position $x \in \Omega$ and time $t \in [0,\infty)$, respectively. Chemotaxis is the movement toward higher concentration of some chemical substance, which is modeled by the flow of u in the direction of the gradient of ρ: $-\alpha\nabla\cdot(u\nabla\rho)$. The term $f(u)$ denotes the proliferation and the reduction due to death of the individuals (we refer to the proliferation and the reduction together simply as growth). In the present paper, the growth function $f(u)$ is assumed to be a Fisher type logistic source:

$$f(u) = au(1-u),$$

where a is a positive constant. The terms $-b\rho$ and cu denote the degradation and the production of the chemical substance, respectively.

It is important to derive a rigorous relationship between the balance of diffusion–chemotaxis–growth (d, α, a) and the spatio-temporal patterns of (u, ρ). Indeed, if there is no growth, that is, $a = 0$, then the system (P) reduces to the classical Keller–Segel system of two parabolic partial differential equations. In the parabolic–parabolic Keller–Segel system, when $N = 1$, for any chemotactic intensity $\alpha > 0$, the solutions exist globally in time without any restriction on the total mass $\|u_0\|_1$, and moreover, exponential attractors have been constructed [2]. On the other hand, when $N = 2$, if $\alpha\|u_0\|_1$ is large, then the value u of some solutions blows up, and among the blow-up solutions, there exists a special one exhibiting a δ-function singularity in a finite time (chemotactic collapse) [3,4]. In contrast, when $N \geq 3$, no

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restriction on α and $\|u_0\|_1$ is necessary for the occurrence of blow-up [5]. For other topics on the Keller–Segel system, we refer the reader to Horstmann's review papers [6,7] and the references therein. In another extreme case, of no chemotaxis, that is, $\alpha = 0$, the first equation reduces to the Fisher equation for which any positive solution u tends to the stable equilibrium $u = 1$ as $t \to \infty$. Therefore, it can be said that (P) provides an interesting problem for studying the dynamics of (u, ρ) driven by the combination of diffusion, chemotaxis and growth.

For the chemotaxis–growth system (P) with $\alpha > 0$ and $a > 0$, the blow-up of the solutions is prevented by the quadratic absorption of the logistic source. In fact, when $N = 2$, the global solutions and exponential attractors are constructed without any restriction on the coefficients in (P) [8]; when $N \geq 3$, if the absorption coefficient is sufficiently large, then the existence of globally bounded solutions is obtained [9]. On the other hand, concerning the blow-up results for the weaker absorption case, to the best of our knowledge, there have been less results; we here only refer to one result, on the system simplified as the parabolic–elliptic type which shows the finite time blow-up under the κ-th order absorption with $1 < \kappa < 3/2 + 1/(2N-2)$ when $N \geq 5$ [10] (see also [11]). For other rigorous results on the chemotaxis–growth systems of parabolic–parabolic type, we refer the reader to [12–15]. In the numerical computation, the Turing patterns induced by the chemotaxis [16,17], which correspond to the experimental results [18,19], have been investigated. Of special interest, Painter and Hillen [20] found that the dependence of spatio-temporal patterns, including a chaotic one, on the chemotactic intensity was numerically studied on the one-dimensional interval. On the other hand, Kurata et al. [17] showed the bifurcation structure of stationary and periodic solutions on a one-dimensional interval by AUTO and stationary patterns with stripe and hexagonal structures and oscillating patterns on the rectangle. It has also been proved that under a periodic boundary condition a stable hexagonal cell structure exists, by the center manifold theory [21]. These results imply a rich structure of asymptotic profiles of solutions as $t \to \infty$. In this sense, studying the stationary solutions in a higher dimensional case is important.

In the present paper, we treat the corresponding stationary problem:

$$\text{(SP)}\quad \begin{cases} d\Delta u - \alpha\nabla\cdot(u\nabla\rho) + au(1-u) = 0 & \text{in } \Omega, \\ \Delta\rho - b\rho + cu = 0 & \text{in } \Omega, \\ \dfrac{\partial u}{\partial\nu} = \dfrac{\partial\rho}{\partial\nu} = 0 & \text{on } \partial\Omega, \\ u \geq 0, \quad \rho \geq 0 & \text{in } \Omega. \end{cases}$$

We are interested in nonconstant solutions of (SP) because these solutions have the possibility of providing information on the stationary patterns. Thanks to the result by Tello and Winkler [22], we know that (SP) has no nonconstant solution if advection $\alpha > 0$ is sufficiently small, while the local branch of nonconstant solutions bifurcates from the unique positive constant solution $(u, \rho) = (1, c/b)$ at each bifurcation point α_j for $\alpha_j \to \infty$ (see Theorem 2.2 for a formal statement). Motivated by these existence results [16,17,22] of nonconstant solutions, the present paper focuses on the following topics:

- Study the asymptotic behavior of nonconstant solutions of (SP) as $\alpha \to \infty$.
- Prove the existence of the stationary stripe and hexagonal patterns in the special case when Ω is a rectangle.
- Show numerical results for several patterns.

The organization of this paper is as follows. In Section 2, we introduce the existence result [22] of nonconstant solutions of (SP). In Section 3, we discuss the asymptotic behavior of nonconstant solutions as $\alpha \to \infty$. Our result can be stated as that any nonconstant solution (u_α, ρ_α) of (SP) approaches $(1, c/b)$ or $(0, 0)$ in the topology of $L^2(\Omega) \times H^2(\Omega)$ as $\alpha \to \infty$ by passing to a subsequence if necessary. In Sections 4–8, we discuss (SP) in the special case when Ω is a rectangle. Our main tool in the analysis is the bifurcation theorem by Crandall and Rabinowitz [23]. In the application of the theorem, we regard α as a bifurcation parameter. In Section 4, we find infinitely many degenerate (bifurcation) points on the constant positive solution set (trivial branch)

$$\Pi := \{(u, \rho, \alpha) = (1, c/b, \alpha) : \alpha > 0\} \subset H^2_\nu(\Omega) \times H^2_\nu(\Omega) \times \boldsymbol{R},$$

where the Hilbert space $H^2_\nu(\Omega)$ is defined by $H^2_\nu(\Omega) = \{u \in H^2(\Omega) : (\partial u)/(\partial\nu) = 0 \text{ on } \partial\Omega\}$. In Sections 5–7, we obtain the local branches of the nonconstant positive solutions, which bifurcate from the bifurcation points on Π. Furthermore, we investigate the behavior of the local branches near the bifurcation point. In order to show the existence of two-dimensional patterns, stripe, rectangular and hexagonal solutions, we introduce the function subspace, for example (6.2), following the method of Nishida et al. [24] for the bifurcation problem in hydrodynamics. In Section 8, we exhibit several numerical patterns with respect to two parameters (a, α) and compare these numerical results with the one from the bifurcation analysis. Finally, we describe our analytic and numeric results in terms of pattern formations and show two oscillating patterns on the rectangle.

Throughout the paper, the usual norms of the spaces $L^p(\Omega)$ for $p \in [1, \infty)$ and $C(\overline{\Omega})$ are defined by $\|u\|_p := (\int_\Omega |u|^p)^{1/p}$ and $\|u\|_\infty := \max_{\overline{\Omega}} |u|$.

2. Known results

In this section, we introduce the known results by Tello and Winkler [22] for the a priori estimates and the existence of nonconstant solutions of (SP). We say that (u, ρ) is a solution if u and ρ are nonnegative functions such that $(u, \rho) \in W^{1,1}(\Omega) \times W^{1,1}(\Omega)$ and $u\nabla\rho \in L^1(\Omega)$ and (u, ρ) satisfies (SP) in the following weak sense:

$$d\int_\Omega \nabla u\cdot\nabla\varphi - \alpha\int_\Omega u\nabla\rho\cdot\nabla\varphi = a\int_\Omega u(1-u)\varphi,$$

$$\int_\Omega \nabla\rho\cdot\nabla\psi + b\int_\Omega \rho\psi = c\int_\Omega u\psi$$

for all $(\varphi, \psi) \in [C^2(\Omega)\cap C(\overline{\Omega})] \times [C^2(\Omega)\cap C(\overline{\Omega})]$.

First we introduce the regularity and a priori estimate for solutions by Tello and Winkler [22].

Lemma 2.1 *([22, Lemma 4.1]). Let (u, ρ) be any solution of (SP). Then (u, ρ) is bounded, satisfies $u, \rho \in C^2(\Omega)\cap C(\overline{\Omega})$ and*

$$\exp\left\{-\alpha\left(\max_{x\in\Omega}\rho(x) - \min_{x\in\Omega}\rho(x)\right)\right\} \leq u(x) \leq \exp\left\{\alpha\left(\max_{x\in\Omega}\rho(x) - \min_{x\in\Omega}\rho(x)\right)\right\}$$

for all $x \in \Omega$.

In fact, Tello and Winkler [22, Lemma 4.1] have obtained Lemma 2.1 for a general class of reaction–diffusion–advection systems (even in the higher dimensional case) including (SP).

Furthermore, Tello and Winkler [22, Theorem 4.3 and Corollary 5.2] obtained infinitely many local branches of nonconstant solutions bifurcating from the positive constant solution and showed that there exists no nonconstant solution when α is small. It is possible to verify that their results can be stated as the following theorem with the aid of the index formula in the Leray–Schauder degree theory.

Theorem 2.2 ([22]). *There exists a positive number* $\underline{\alpha}$ *and a positive sequence* $\{\alpha_j\}$ *with*

$$0 < \underline{\alpha} < \alpha_1 \le \alpha_2 \le \cdots \le \alpha_j \le \cdots \to \infty \quad (j \to \infty)$$

such that

(i) *if* $\alpha \in (0, \underline{\alpha}]$, *then (SP) has no nonconstant solution,*
(ii) *if* $\alpha \in (\alpha_j, \alpha_{j+1})$, $\alpha_j \ne \alpha_{j+1}$ *and* j *is odd, then (SP) possesses at least one nonconstant positive solution.*

Thanks to (i) of Theorem 2.2, we know that (SP) has no positive solution except for constant solution $(u, \rho) = (1, c/b)$ in the small advection case. On the other hand, (ii) of Theorem 2.2 gives sufficient conditions on α for the existence of nonconstant solutions of (SP). See the discussion including (4.9) for the definition of $\{\alpha_j\}$ in the case when Ω is the rectangle (4.1), Additionally, Tello and Winkler [22, Theorem 5.1] proved that any solution (u, v) of (P) with nonnegative initial data $(u_0, v_0) \ne (0, 0)$ approaches the constant solution $(1, c/b)$ as $t \to \infty$ when α is small. This result implies that any nonconstant stationary pattern cannot appear in the small advection case. In terms of pattern formation, it is natural to be interested in the existence and stability of solutions of (SP) (or (P)) in the case when α is not small.

3. Asymptotic behavior of solutions as $\alpha \to \infty$

In this section, we study the asymptotic behavior of positive solutions of (SP) as $\alpha \to \infty$. For the first step of the asymptotic analysis, we derive the a priori estimates for every positive solution (u, ρ) independent of (d, a, α).

Lemma 3.1. *Let* (u, ρ) *be any positive solution of (SP). Then* u *satisfies*

$$\|u\|_1 = \|u\|_2^2 \le |\Omega| \tag{3.1}$$

and

$$\min_{\overline{\Omega}} u \le 1 \le \max_{\overline{\Omega}} u. \tag{3.2}$$

There exists a positive constant C *independent of* (d, α, a) *such that*

$$\|\rho\|_{H^2} \le C.$$

Proof. By integrating the first equation of (SP), we obtain

$$\int_\Omega u(1-u) = 0.$$

Furthermore, Hölder's inequality implies

$$\int_\Omega u^2 = \int_\Omega u \le |\Omega|^{1/2} \left(\int_\Omega u^2 \right)^{1/2}, \tag{3.3}$$

which immediately yields (3.1). Since

$$\left(\min_{\overline{\Omega}} u \right) \int_\Omega u \le \int_\Omega u^2 \le \left(\max_{\overline{\Omega}} u \right) \int_\Omega u,$$

together with (3.3) we have (3.2).

By applying the elliptic regularity theory (see, e.g., [25,26]) to the second equation of (SP), it follows that $\|\rho\|_{H^2} \le C$ for a positive constant C. □

By making use of Lemma 3.1, we derive the following asymptotic behavior of solutions as $\alpha \to \infty$.

Theorem 3.2. *Let* (u_α, ρ_α) *be any positive solution of (SP) for* $N \le 3$. *Then there exists a subsequence* $\{\alpha_n\}$ *with* $\lim_{n\to\infty} \alpha_n = \infty$ *such that* $(u_n, \rho_n) := (u_{\alpha_n}, \rho_{\alpha_n})$ *satisfies one of the following:*

(i) $\lim_{n\to\infty}(u_n, \rho_n) = (0, 0)$ *in* $L^2(\Omega) \times H^2(\Omega)$,
(ii) $\lim_{n\to\infty}(u_n, \rho_n) = (1, \frac{c}{b})$ *in* $L^2(\Omega) \times H^2(\Omega)$.

Proof. The uniform boundedness of $\{\|u_\alpha\|_2, \|\rho_\alpha\|_{H^2}\}$ for any $\alpha > 0$ from Lemma 3.1 enables us to find a sequence $\{\alpha_n\}$ with $\lim_{n\to\infty} \alpha_n = \infty$ such that $(u_n, \rho_n) := (u_{\alpha_n}, \rho_{\alpha_n})$ satisfies

$$\begin{cases} \lim_{n\to\infty} u_n = u_\infty & \text{weakly in } L^2(\Omega), \\ \lim_{n\to\infty} \rho_n = \rho_\infty & \text{weakly in } H^2(\Omega), \\ \lim_{n\to\infty} \rho_n = \rho_\infty & \text{strongly in } H^1(\Omega) \end{cases} \tag{3.4}$$

for some $(u_\infty, \rho_\infty) \in L^2(\Omega) \times H^2(\Omega)$ and ρ_∞ is a weak solution of

$$-\Delta \rho_\infty + b\rho_\infty = cu_\infty \quad \text{in } \Omega, \qquad \frac{\partial \rho_\infty}{\partial \nu} = 0 \quad \text{on } \partial\Omega. \tag{3.5}$$

Furthermore, (3.1) and (3.4) ensure

$$\|u_\infty\|_2 \le \liminf_{n\to\infty} \|u_n\|_2 \le |\Omega|^{1/2}.$$

We introduce the set

$$\mathcal{S} := \left\{ \varphi \in H^2(\Omega) : \frac{\partial \varphi}{\partial \nu} = 0 \text{ on } \partial\Omega \right\}.$$

Then by multiplying the first equation of (SP) by $\varphi \in \mathcal{S}$ and integrating the resulting expression, we have

$$\frac{d}{\alpha_n} \int_\Omega u_n \Delta\varphi + \int_\Omega u_n \nabla\rho_n \cdot \nabla\varphi + \frac{a}{\alpha_n} \int_\Omega u_n(1-u_n)\varphi = 0. \tag{3.6}$$

As $n \to \infty$, it follows from (3.4) and (3.6) that

$$\int_\Omega u_\infty \nabla\rho_\infty \cdot \nabla\varphi = 0 \quad \text{for any } \varphi \in \mathcal{S}. \tag{3.7}$$

For $\varphi = \rho_\infty$, we see that

$$\int_\Omega u_\infty |\nabla\rho_\infty|^2 = 0. \tag{3.8}$$

First we prove that ρ_∞ is constant. To do so, we show that the Lebesgue measure of the set

$$\mathcal{K} := \{x \in \Omega : |\nabla\rho_\infty| > 0\}$$

is zero. Assume the contrary, that $\mathcal{K}$ possesses a positive measure. Then (3.8) implies $u_\infty = 0$ almost everywhere in $\mathcal{K}$. Therefore, it follows from (3.5) that $\rho_\infty = 0$ in $\mathcal{K}$, which is a contradiction. Then we can show $|\mathcal{K}| = 0$, that is,

$$|\nabla\rho_\infty| = 0 \quad \text{almost everywhere in } \Omega. \tag{3.9}$$

Since $\rho_\infty \in C(\Omega)$ by the Sobolev embedding theorem ($H^2(\Omega) \subset C(\overline{\Omega})$ if $N \le 3$), one can deduce from (3.9) that ρ_∞ is a nonnegative constant in Ω. Then (3.5) implies that u_∞ equals a nonnegative constant almost everywhere in Ω. Therefore, by integrating the first equation of (SP) and letting $n \to \infty$ in the resulting expression, one can see that

$$\int_\Omega u_n^2 = \int_\Omega u_n \to u_\infty|\Omega| \quad \text{as } n \to \infty. \tag{3.10}$$

Hence, (3.10) ensures that $\{u_n\}$ contains a subsequence $\{u_{n'}\}$ which converges to u_∞ almost everywhere in Ω as $n' \to \infty$. Since

$$\lim_{n\to\infty} u_{n'}^2 = u_\infty^2 \quad \text{almost everywhere in } \Omega,$$

(3.10) and the Lebesgue convergence theorem imply

$$\int_\Omega u_{n'}^2 \to u_\infty^2|\Omega| \quad \text{as } n' \to \infty. \tag{3.11}$$

Therefore, by (3.10) and (3.11), we can deduce that $u_\infty(1-u_\infty) = 0$ almost everywhere in Ω. Recall that u_∞ is a nonnegative constant almost everywhere in Ω. If $u_\infty = 0$ almost everywhere in Ω, then we have $\rho_\infty = 0$ by (3.5), whereas if $u_\infty = 1$ almost everywhere in Ω, then $\rho_\infty = c/b$. It follows from (3.4) and (3.11) that

$$\lim_{n'\to\infty} u_{n'} = u_\infty \quad \text{strongly in } L^2(\Omega).$$

Thus the proof of Theorem 3.2 is complete. □

Remark 3.1. Since $(u, \rho) = (1, c/b)$ is a constant solution, the existence of the sequence $\{(u_n, \rho_n)\}$ satisfying (ii) of Theorem 3.2 is trivial. On the other hand, if there exists a sequence $\{(u_n, \rho_n)\}$ satisfying (i) of Theorem 3.2, then u_n cannot converge to zero in the topology of $C(\overline{\Omega})$, because $\max_{\overline{\Omega}} u_n \geq 1$ for all $n \in \boldsymbol{N}$ by (3.2). It is important to prove the existence of sequence $\{(u_n, \rho_n)\}$ satisfying (i) of Theorem 3.2 and this is our future problem. We note that Dancer et al. [27] constructed peak solutions of a related elliptic system under the Dirichlet boundary conditions and another scaling for parameters.

4. Degeneracy points on the constant solution set

Hereinafter, we study (SP) in the special case when the habitat Ω is a rectangle;

$$\Omega = \left(0, \frac{\pi}{l}\right) \times \left(0, \frac{\pi}{\sqrt{3}l}\right) \quad (l > 0). \tag{4.1}$$

In this domain, we show the existence of *stripe*, *rectangular* and *hexagonal* solutions bifurcating from the positive constant solution $(u, \rho) = (1, c/b)$. For the framework of the bifurcation argument, we introduce two Hilbert spaces X and Y defined by

$$X = H^2_\nu(\Omega) \times H^2_\nu(\Omega), \qquad Y = L^2(\Omega) \times L^2(\Omega).$$

By regarding α as the bifurcation parameter, we can find the bifurcation points on the trivial branch $\Pi = \{(1, c/b, \alpha) : \alpha \geq 0\} \subset X \times \boldsymbol{R}$. Although the search for the bifurcation points is essentially the same as in the proof of Theorem 4.3 in [22], we will describe the process in this section to refer to it in a later section.

For the application of the local bifurcation theory by Crandall and Rabinowitz [23], we define the operator $F : X \times \boldsymbol{R} \to Y$ associated with (SP) by

$$F(u, \rho, \alpha) = \begin{pmatrix} d\Delta u - \alpha\nabla(u\nabla\rho) + au(1-u) \\ \Delta\rho - b\rho + cu \end{pmatrix}. \tag{4.2}$$

By virtue of the implicit function theorem, the linearized operator $F_{(u,\rho)}(\alpha)$ around $(u, \rho) = (1, c/b)$ is defined by

$$F_{(u,\rho)}(\alpha)\begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} d\Delta h - \alpha\Delta k - ah \\ \Delta k + ch - bk \end{pmatrix} \tag{4.3}$$

and must be degenerate at any bifurcation point, that is, we seek α such that $F_{(u,\rho)}(\alpha)$ has a zero eigenvalue. To do so, we consider the linear elliptic boundary value problem:

$$\text{(L)} \begin{cases} d\Delta h - \alpha\Delta k - ah = 0 & \text{in } \Omega, \\ \Delta k + ch - bk = 0 & \text{in } \Omega, \\ \dfrac{\partial h}{\partial \nu} = \dfrac{\partial k}{\partial \nu} = 0 & \text{on } \partial\Omega. \end{cases}$$

Let a solution (h, k) of (L) be represented by the Fourier expansion formula:

$$\begin{aligned} h(x, y) &= \sum_{m,n=0}^{\infty} h_{mn}\phi_m(x)\psi_n(y), \\ k(x, y) &= \sum_{m,n=0}^{\infty} k_{mn}\phi_m(x)\psi_n(y), \end{aligned} \tag{4.4}$$

where

$$\phi_m(x) := \cos(lmx), \qquad \psi_n(y) := \cos(\sqrt{3}ln\,y). \tag{4.5}$$

Substituting (4.4) into (L), we show that (h_{mn}, k_{mn}) satisfies

$$\begin{cases} \displaystyle\sum_{m,n=0}^{\infty} [\{dl^2(m^2+3n^2)+a\}h_{mn} - \alpha l^2(m^2+3n^2)k_{mn}] \\ \quad \phi_m(x)\psi_n(y) = 0, \\ \displaystyle\sum_{m,n=0}^{\infty} [ch_{mn} - \{l^2(m^2+3n^2)+b\}k_{mn}]\phi_m(x)\psi_n(y) = 0. \end{cases} \tag{4.6}$$

Since $\{\phi_m(x)\psi_n(y)\}_{m,n=0}^{\infty}$ forms a complete orthogonal base of X, (4.6) can be reduced to the algebraic equations

$$\begin{pmatrix} dl^2(m^2+3n^2)+a & -\alpha l^2(m^2+3n^2) \\ c & -\{l^2(m^2+3n^2)+b\} \end{pmatrix}\begin{pmatrix} h_{mn} \\ k_{mn} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{4.7}$$

for all $m, n \in \boldsymbol{N} \cup \{0\}$. Therefore, (L) has nontrivial solutions if and only if (4.7) has nontrivial solutions for some (m, n), that is,

$$\begin{aligned} &\begin{vmatrix} dl^2(m^2+3n^2)+a & -\alpha l^2(m^2+3n^2) \\ c & -\{l^2(m^2+3n^2)+b\} \end{vmatrix} \\ &= -\{dl^2(m^2+3n^2)+a\}\{l^2(m^2+3n^2)+b\} + \alpha cl^2(m^2+3n^2) \\ &= 0. \end{aligned}$$

Therefore, $\operatorname{Ker} F_{(u,\rho)}(\alpha)$ is nontrivial if and only if

$$\alpha = \frac{\{dl^2(m^2+3n^2)+a\}\{l^2(m^2+3n^2)+b\}}{cl^2(m^2+3n^2)} \; (=: \alpha(m,n)). \tag{4.8}$$

Hence, $\alpha(m, n) > 0$, which implies that the positive constant solution is non-degenerate without the advection term.

For example, in the simplest case when

$$\alpha(1, 0) = \frac{(dl^2+a)\,(l^2+b)}{cl^2},$$

we know that $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(1, 0)) = 1$. However, we remark that $(m, n) \mapsto \alpha(m, n)$ is not a one-to-one correspondence. For example,

$$\alpha(2, 0) = \alpha(1, 1) = \frac{(4dl^2+a)\,(4l^2+b)}{4cl^2} \tag{4.9}$$

or

$$\alpha(3, 1) = \alpha(0, 2) = \frac{(12dl^2+a)\,(12l^2+b)}{12cl^2}.$$

It is verified that $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m, n)) = 2$ if $(m, n) = (2, 0)$, $(1, 1)$ or $(3, 1)$, $(0, 2)$. Furthermore,

$$\alpha(1, 3) = \alpha(5, 1) = \alpha(4, 2) = \frac{(28dl^2+a)\,(28l^2+b)}{28cl^2},$$

which implies $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m, n)) = 3$ for $(m, n) = (1, 3)$, $(5, 1)$, $(4, 2)$. In the case when Ω is the rectangle, the sequence $\{\alpha_j\}$ in Theorem 2.2 is determined by putting the $\{\alpha(m, n)\}$ in order (counting multiplicity).

We study the one-dimensional degenerate cases, that is, $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m, n)) = 1$, in Sections 5 and 7, and the special case (4.9) as a simple example of the two-dimensional degenerate case in Section 6.

5. Pitchfork bifurcation branch of stripe or rectangular patterns

In the case of $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m, n)) = 1$, the bifurcation theory of [23] is applicable to constructing the local branch of nonconstant solutions, which bifurcate from the trivial branch Π. Furthermore, we prove that this bifurcation diagram of nonconstant solutions is *pitchfork* type.

Theorem 5.1. *If* $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m,n)) = 1$, *there exists a positive constant* δ *such that nonconstant solutions of (SP) near* $(u,\rho,\alpha) = (1, c/b, \alpha(m,n)) \in X \times \boldsymbol{R}$ *can be represented by*

$$\begin{pmatrix} u(s) \\ \rho(s) \\ \alpha(s) \end{pmatrix} = \begin{pmatrix} 1 \\ c/b \\ \alpha(m,n) \end{pmatrix} + s \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \\ 0 \end{pmatrix} + s^2 \begin{pmatrix} \tilde{u}(s) \\ \tilde{\rho}(s) \\ \tilde{\alpha}(s) \end{pmatrix} \quad (5.1)$$

for all $s \in [-\delta, \delta]$. *Here* $\phi_m(x)$ *and* $\psi_n(y)$ *are defined in (4.5),* $(\tilde{u}(s), \tilde{\rho}(s), \tilde{\alpha}(s)) \in X \times \boldsymbol{R}$ *is a smooth function of* s, *and moreover,*

$$k_{mn} = \frac{c}{l^2(m^2+3n^2)+b}, \quad (5.2)$$

which is obtained from (4.7) with the normalization $h_{mn} = 1$.

Proof. In view of $\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m,n)) = 1$ and (4.7), we have

$$\operatorname{Ker} F_{(u,\rho)}(\alpha(m,n)) = \operatorname{span}\left\{ \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} \right\} \quad (5.3)$$

for the positive number k_{mn} defined in (5.2). In order to apply the local bifurcation theorem of [23] at $(u,\rho,\alpha) = (1, c/b, \alpha(m,n))$, we need to verify

$$F_{(u,\rho),\alpha}(\alpha(m,n)) \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} \notin \operatorname{Range} F_{(u,\rho)}(\alpha(m,n)), \quad (5.4)$$

where $F_{(u,\rho),\alpha}$ denotes the derivative of $F_{(u,\rho)}$ with respect to α, that is,

$$F_{(u,\rho),\alpha}(\alpha) \begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} -\Delta k \\ 0 \end{pmatrix}.$$

Therefore, we obtain

$$F_{(u,\rho),\alpha}(\alpha(m,n)) \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} = \begin{pmatrix} -k_{mn}\Delta(\phi_m(x)\psi_n(y)) \\ 0 \end{pmatrix}$$
$$= \begin{pmatrix} k_{mn} l^2(m^2+3n^2)\phi_m(x)\psi_n(y) \\ 0 \end{pmatrix}. \quad (5.5)$$

On the other hand, the adjoint operator L^*_{mn} of $F_{(u,\rho)}(\alpha(m,n))$ is defined by

$$L^*_{mn}\begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} d\Delta h - ah + ck \\ -\alpha(m,n)\Delta h + \Delta k - bk \end{pmatrix}.$$

Furthermore, following the same procedure for L^*_{mn} as in the previous section, one can see that

$$\operatorname{Ker} L^*_{mn} = \operatorname{span}\left\{ \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k^*_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} \right\}, \quad (5.6)$$

where

$$k^*_{mn} = \frac{dl^2(m^2+3n^2)+a}{c}.$$

From the Fredholm alternative theorem, we know that

$$\operatorname{Range} F_{(u,\rho)}(\alpha(m,n)) = (\operatorname{Ker} L^*_{mn})^{\perp}. \quad (5.7)$$

By taking the L^2 inner product between the right-hand side of (5.5) and the base of Ker L^*_{mn}, we have

$$\int_\Omega k_{mn}\phi_m^2(x)\psi_n^2(y)dxdy > 0. \quad (5.8)$$

The positivity of (5.8) implies

$$F_{(u,\rho),\alpha}(\alpha(m,n)) \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} \notin (\operatorname{Ker} L^*_{mn})^{\perp},$$

which immediately yields (5.4), because of (5.7). Consequently, from the local bifurcation theorem, there exists a positive constant δ such that all nonconstant solutions of (SP) near $(u,\rho,\alpha) = (1, c/b, \alpha(m,n)) \in X \times \boldsymbol{R}$ can be represented by

$$\begin{pmatrix} u(s) \\ \rho(s) \end{pmatrix} = \begin{pmatrix} 1 \\ c/b \end{pmatrix} + s \begin{pmatrix} \phi_m(x)\psi_n(y) \\ k_{mn}\phi_m(x)\psi_n(y) \end{pmatrix} + s^2 \begin{pmatrix} \tilde{u}(s) \\ \tilde{\rho}(s) \end{pmatrix}$$

and

$$\alpha(s) = \alpha(m,n) + s\beta(s)$$

for $s \in [-\delta, \delta]$.

In order to show (5.1), we have only to prove

$$\beta(0) = 0. \quad (5.9)$$

For simplicity, we set

$$\Phi(x,y) = \phi_m(x)\psi_n(y). \quad (5.10)$$

It should be noted that

$$\langle \Phi, \tilde{u}(s)\rangle := \int_\Omega \Phi(x,y)\,\tilde{u}(x,y,s)\,dxdy = 0,$$
$$\langle \Phi, \tilde{\rho}(s)\rangle := \int_\Omega \Phi(x,y)\,\tilde{\rho}(x,y,s)\,dxdy = 0 \quad (5.11)$$

for $s \in [-\delta, \delta]$. Henceforth, let λ be defined by

$$-\Delta\Phi = \lambda\Phi \quad \text{in } \Omega. \quad (5.12)$$

By differentiating the first equation of (SP) twice with respect to s, we obtain

$$\begin{aligned} &d\Delta u''(s) - \alpha''(s)\nabla\{u(s)\nabla\rho(s)\} \\ &\quad - 2\alpha'(s)\nabla\{u'(s)\nabla\rho(s) + u(s)\nabla\rho'(s)\} \\ &\quad - \alpha(s)\nabla\{u''(s)\nabla\rho(s) + 2u'(s)\nabla\rho'(s) + u(s)\nabla\rho''(s)\} \\ &\quad + au''(s)\{1-2u(s)\} - 2au'(s)^2 = 0. \end{aligned} \quad (5.13)$$

As $s = 0$ in (5.13), we have

$$\begin{aligned} &d\Delta\tilde{u}(0) + k_{mn}\lambda\beta(0)\Phi - \alpha(m,n)\Delta\tilde{\rho}(0) \\ &\quad = a(\tilde{u}(0) + \Phi^2) + \alpha(m,n)k_{mn}(|\nabla\Phi|^2 - \lambda\Phi^2). \end{aligned} \quad (5.14)$$

Using integration by parts, we can observe that

$\langle \Delta\tilde{u}(0), \Phi\rangle = \langle \tilde{u}(0), \Delta\Phi\rangle = -\lambda\langle \tilde{u}(0), \Phi\rangle = 0$ and $\langle \Delta\tilde{\rho}(0), \Phi\rangle = 0$.

Therefore, by taking the L^2 inner product of (5.14) with Φ, we obtain

$$\begin{aligned} k_{mn}\lambda\|\Phi\|_2^2\beta(0) &= (a - \lambda\alpha(m,n)k_{mn})\langle \Phi, \Phi^2\rangle \\ &\quad + \alpha(m,n)k_{mn}\langle \Phi, |\nabla\Phi|^2\rangle. \end{aligned} \quad (5.15)$$

By (5.10), we see that

$$\langle \Phi, \Phi^2\rangle = \int_0^{\pi/l}\phi_m^3(x)\,dx\int_0^{\pi/\sqrt{3}l}\psi_n^3(y)\,dy = 0 \quad (5.16)$$

and

$$\begin{aligned} \langle \Phi, |\nabla\Phi|^2\rangle &= \int_\Omega \phi_m(x)\psi_n(y)l^2\{m^2\sin^2(lmx)\psi_n^2(y) \\ &\qquad + 3n^2\phi_m^2(x)\sin^2(\sqrt{3}\ln y)\}\,dxdy \\ &= l^2m^2\int_0^{\pi/l}\phi_m(x)\sin^2(lmx)\,dx \\ &\qquad \times \int_0^{\pi/\sqrt{3}l}\psi_n^3(y)\,dy + 3l^2n^2\int_0^{\pi/l}\phi_m^3(x)\,dx \\ &\qquad \times \int_0^{\pi/\sqrt{3}l}\psi_n(y)\sin^2(\sqrt{3}ln\,y)\,dy = 0. \end{aligned} \quad (5.17)$$

Then by substituting (5.16) and (5.17) into (5.15), we obtain $\beta(0) = 0$, which immediately yields (5.1). This completes the proof of Theorem 5.1. □

The parameterization (5.1) implies that if $mn = 0$, the local branch of the *stripe* solutions bifurcates from the constant solution $(1, c/b)$ at $\alpha = \alpha(m, n)$, whereas if $mn \neq 0$, the local branch of the *rectangular* solutions bifurcates from the constant solution $(1, c/b)$ at $\alpha = \alpha(m, n)$. Moreover, (5.1) implies that the bifurcation diagram is pitchfork type (recall (5.9)) and

$$\rho(s) - \frac{c}{b} = k_{mn}(u(s) - 1) + O(s^2) \quad \text{for all } |s| \leq \delta. \tag{5.18}$$

Since $k_{mn} > 0$ from (5.2),

(5.18) implies that u and ρ have a same characteristic with respect to the profiles.

6. Crossing bifurcation branch of hexagonal patterns

6.1. Local bifurcation branch of hexagonal patterns

In this section, we study the case (4.9) as a typical case when

$$\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(m, n)) = 2.$$

Among other things, we will show that a local branch of the *hexagonal* solutions bifurcates from the trivial branch Π at $\alpha = \alpha(2, 0) = \alpha(1, 1)$. It is easy to show that

$$\operatorname{Ker} F_{(u,\rho)}(\alpha(2, 0)) = \operatorname{Ker} F_{(u,\rho)}(\alpha(1, 1))$$
$$= \operatorname{span}\left\{ \begin{pmatrix} \phi_2(x) \\ k_{20}\,\phi_2(x) \end{pmatrix}, \begin{pmatrix} \phi_1(x)\,\psi_1(y) \\ k_{11}\,\phi_1(x)\,\psi_1(y) \end{pmatrix} \right\} \tag{6.1}$$

for the positive constants k_{20} and k_{11} introduced by (5.2). In this case, we cannot apply the local bifurcation theory in X. Therefore, following the idea of Nishida et al. [24] for the bifurcation problem in hydrodynamics, we introduce the closed subspace H^2_{hexa} of $H^2_\nu(\Omega)$, defined by

$$H^2_{\text{hexa}} = \left\{ v(x, y) = \sum_{m+n=\text{even}}^{\infty} \beta_{mn} \left(\phi_m(x)\psi_n(y) + \cos\frac{l(m-3n)x}{2} \cos\frac{\sqrt{3}l(m+n)y}{2} + \cos\frac{l(m+3n)x}{2} \cos\frac{\sqrt{3}l(m-n)y}{2} \right) : \sum_{m+n=\text{even}}^{\infty} l^4(m^2+3n^2)^2\beta_{mn}^2 < \infty \right\}. \tag{6.2}$$

We remark that an element of H^2_{hexa} is invariant with respect to the $2\pi/3$-rotation. In view of (6.2), by letting $(m, n) = (2, 0)$ in

$$\phi_m(x)\psi_n(y) + \cos\frac{l(m-3n)x}{2} \cos\frac{\sqrt{3}l(m+n)y}{2} + \cos\frac{l(m+3n)x}{2} \cos\frac{\sqrt{3}l(m-n)y}{2},$$

we see that

$$\phi_2(x)\psi_0(y) + \cos(lx)\cos(\sqrt{3}ly) + \cos(lx)\cos(\sqrt{3}ly)$$
$$= \phi_2(x)\psi_0(y) + 2\phi_1(x)\psi_1(y).$$

Consequently, setting

$$\beta_{mn} = \begin{cases} 1 & \text{if } (m, n) = (2, 0), \\ 0 & \text{otherwise} \end{cases}$$

in (6.2), we have $\phi_2(x)\psi_0(y) + 2\phi_1(x)\psi_1(y) \in H^2_{\text{hexa}}$. This implies that a *hexagonal* pattern is represented by $\phi_2(x)\psi_0(y) + 2\phi_1(x)\psi_1(y)$. As in the case of (6.1), $\alpha(2, 0)\ (= \alpha(1, 1))$ is a double eigenvalue in the sense that

$$\dim \operatorname{Ker} F_{(u,\rho)}(\alpha(1, 1)) = \dim \operatorname{Ker} F_{(u,\rho)}(\alpha(2, 0)) = 2.$$

Next we introduce the operator $\tilde{F}$ by a restriction of F (defined by (4.2)) to the domain of $H^2_{\text{hexa}} \times H^2_{\text{hexa}} \times \boldsymbol{R}$, namely, $\tilde{F} : H^2_{\text{hexa}} \times H^2_{\text{hexa}} \times \boldsymbol{R} \to Y$. Therefore it follows that

$$\operatorname{Ker} \tilde{F}_{(u,\rho)}(\alpha(1, 1)) = \operatorname{Ker} \tilde{F}_{(u,\rho)}(\alpha(2, 0))$$
$$= \operatorname{span}\left\{ \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\,\psi_1(y) \\ k_{20}(\phi_2(x) + 2\phi_1(x)\,\psi_1(y)) \end{pmatrix} \right\},$$

and

$$\dim \operatorname{Ker} \tilde{F}_{(u,\rho)}(\alpha(1, 1)) = \dim \operatorname{Ker} \tilde{F}_{(u,\rho)}(\alpha(2, 0)) = 1. \tag{6.3}$$

For the restriction operator $\tilde{F}$, we can apply the local bifurcation theory of [23] to obtain the following local bifurcation branch of the *hexagonal* pattern of (SP).

Theorem 6.1. *When $\alpha = \alpha(2, 0)\ (= \alpha(1, 1))$, there exist a positive constant δ and a neighborhood $\mathcal{O}_{\text{hexa}}$ of $(u, \rho, \alpha) = (1, c/b, \alpha(2, 0))$ in $H^2_{\text{hexa}} \times H^2_{\text{hexa}} \times \boldsymbol{R}$ such that nonconstant solutions of (SP) contained in $\mathcal{O}_{\text{hexa}}$ can be represented by*

$$\begin{pmatrix} u(s) \\ \rho(s) \end{pmatrix} = \begin{pmatrix} 1 \\ c/b \end{pmatrix} + s \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\psi_1(y) \\ k_{20}\,\{\phi_2(x) + 2\phi_1(x)\psi_1(y)\} \end{pmatrix} + s^2 \begin{pmatrix} \tilde{u}(s) \\ \tilde{\rho}(s) \end{pmatrix} \tag{6.4}$$

and

$$\alpha(s) = \alpha(2, 0) + s\beta(s) \tag{6.5}$$

for $s \in [-\delta, \delta]$. Here $\phi_m(x)$ and $\psi_n(y)$ are defined in (4.5), k_{20} is the positive constant in (5.2) with $(m, n) = (2, 0)$, and $(\tilde{u}(s), \tilde{\rho}(s), \beta(s)) \in H^2_{\text{hexa}} \times H^2_{\text{hexa}} \times \boldsymbol{R}$ is a smooth function of s.

Proof. Thanks to (6.3), we only have to verify

$$\tilde{F}_{(u,\rho),\alpha}(\alpha(2, 0)) \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\psi_1(y) \\ k_{20}(\phi_2(x) + 2\phi_1(x)\psi_1(y)) \end{pmatrix}$$
$$\notin \operatorname{Range} \tilde{F}_{(u,\rho)}(\alpha(2, 0)). \tag{6.6}$$

In view of (4.5) and (5.4), one can easily obtain

$$\tilde{F}_{(u,\rho),\alpha}(\alpha(2, 0)) \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\psi_1(y) \\ k_{20}\{\phi_2(x) + 2\phi_1(x)\psi_1(y)\} \end{pmatrix}$$
$$= \begin{pmatrix} -k_{20}\Delta(\phi_2(x) + 2\phi_1(x)\psi_1(y)) \\ 0 \end{pmatrix}$$
$$= \begin{pmatrix} 4k_{20}l^2\{\phi_2(x) + \phi_1(x)\psi_1(y)\} \\ 0 \end{pmatrix}.$$

By the Fredholm alternative theorem, the adjoint operator $\tilde{L}^*$ of $\tilde{F}_{(u,\rho)}(\alpha(2, 0))$ satisfies

$$\operatorname{Range} \tilde{F}_{(u,\rho)}(\alpha(2, 0)) = (\operatorname{Ker} \tilde{L}^*)^{\perp}. \tag{6.7}$$

Here $\tilde{L}^* : H^2_{\text{hexa}} \times H^2_{\text{hexa}} \to Y$ is defined by

$$\tilde{L}^* \begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} d\Delta h - ah + ck \\ -\alpha(2, 0)\Delta h + \Delta k - bk \end{pmatrix}.$$

After some computations, one can obtain

$$\operatorname{Ker} \tilde{L}^* = \operatorname{span}\left\{ \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\,\psi_1(y) \\ k^*_{20}\{\phi_2(x) + 2\phi_1(x)\,\psi_1(y)\} \end{pmatrix} \right\}, \tag{6.8}$$

where k_{20}^* is the positive number defined by (5.7) with $(m,n) = (2,0)$. By taking the L^2 inner product of the right-hand sides of (6.7) and the base of Ker $\tilde{L}^*$, we have

$$\int_\Omega \{\phi_2(x) + 2\phi_1(x)\psi_1(y)\}^2 dxdy > 0,$$

which implies

$$\tilde{F}_{(u,\rho),\alpha}(\alpha(2,0)) \begin{pmatrix} \phi_2(x) + 2\phi_1(x)\psi_1(y) \\ k_{20}\{\phi_2(x) + 2\phi_1(x)\psi_1(y)\} \end{pmatrix} \notin (\mathrm{Ker}\,\tilde{L}^*)^\perp. \tag{6.9}$$

Then we can show (6.6) from (6.7) and (6.9). Consequently, we can apply the local bifurcation theory to $\tilde{F}$ at $(u,\rho,\alpha) = (1, c/b, \alpha(2,0))$ and obtain (6.4) and (6.5). Thus the proof of Theorem 6.1 is complete. □

6.2. Crossing bifurcation of hexagonal patterns

We now remark that the bifurcation diagram of the hexagonal solutions represented by (6.4) and (6.5) is generically *transversal* with respect to the trivial branch Π, which is different from the pitchfork bifurcation of the stripe and rectangle types (Theorem 5.1). Actually, we get the following expression for $\beta(0)$.

Theorem 6.2. *Let $\beta(s)$ be the function obtained in (6.5). Then*

$$\beta(0) = \frac{4l^2 + b}{2c}\left(\frac{a}{4l^2} - d\right). \tag{6.10}$$

Proof. In view of (6.4) and (6.5), we set

$$\Phi(x,y) = \phi_2(x) + 2\phi_1(x)\,\psi_1(y). \tag{6.11}$$

Then by a similar manner to (5.13)–(5.15), we obtain

$$k_{20}\lambda\|\Phi\|_2^2\beta(0) = (a - \lambda\alpha(2,0)k_{20})\langle\Phi, \Phi^2\rangle + \alpha(2,0)k_{20}\langle\Phi, |\nabla\Phi|^2\rangle. \tag{6.12}$$

Note that

$$\langle\Phi,\Phi^2\rangle = \frac{\sqrt{3}\pi^2}{2l^2}, \qquad \langle\Phi, |\nabla\Phi|^2\rangle = \sqrt{3}\pi^2, \tag{6.13}$$

and

$$\|\Phi\|_2^2 = \frac{\sqrt{3}\pi^2}{2l^2}, \qquad \lambda = 4l^2, \qquad k_{20} = \frac{c}{4l^2 + b}. \tag{6.14}$$

Then by substituting (6.13) and (6.14) into (6.12), we obtain (6.10). Thus the proof of Theorem 6.2 is complete. □

Therefore it follows from (5.15)–(5.17), (6.10) and (6.13) that the type of the bifurcation diagram, pitchfork or transversal, near the bifurcation points depends on the value of the inner products between the eigenfunctions.

7. Direction of the pitchfork bifurcation branch

7.1. Exact expression for $\tilde{\alpha}(0)$ of Theorem 5.1

In this section, we investigate the direction of the pitchfork bifurcation of the local branch obtained in Theorem 5.1, which implies that the local bifurcation branch of the nonconstant solutions can be expressed by

$$\begin{pmatrix} u(x,y,s) \\ \rho(x,y,s) \\ \alpha(s) \end{pmatrix} = \begin{pmatrix} 1 \\ c/b \\ \alpha(m,n) \end{pmatrix} + s\begin{pmatrix} \Phi(x,y) \\ k_{mn}\Phi(x,y) \\ 0 \end{pmatrix} + s^2\begin{pmatrix} \tilde{u}(x,y,s) \\ \tilde{\rho}(x,y,s) \\ \tilde{\alpha}(s) \end{pmatrix} \tag{7.1}$$

for all $s \in [-\delta, \delta]$. Here $\alpha(m,n)$ is given in (4.8), k_{mn} is the positive constant defined by (5.2) and $\Phi(x,y) = \phi_m(x)\psi_n(y)$. The aim of this section is to obtain an explicit expression for $\tilde{\alpha}(0)$, which determines the direction of the branch (7.1) at the bifurcation point $(1, c/b, \alpha(m,n))$ with respect to the trivial branch Π. If $\tilde{\alpha}(0) > 0$, the pitchfork bifurcation (7.1) is usually called *supercritical*, and if $\tilde{\alpha}(0) < 0$, it is called *subcritical*. The next theorem asserts that $\tilde{\alpha}(0)$ can be expressed by four L^2 inner products:

$$(A,B,C,D) := (\langle\tilde{u}(0), \Phi^2\rangle, \langle\tilde{u}(0), |\nabla\Phi|^2\rangle, \langle\tilde{\rho}(0), \Phi^2\rangle, \langle\tilde{\rho}(0), |\nabla\Phi|^2\rangle). \tag{7.2}$$

Theorem 7.1. *The function $\tilde{\alpha}(s)$ in (7.1) satisfies*

$$\lambda k_{mn}\tilde{\alpha}(0)\|\Phi\|_2^2 = 2aA - \alpha(m,n)\,(k_{mn}B + \lambda C - D). \tag{7.3}$$

Concerning (A,B,C,D), we obtain the following lemma.

Lemma 7.2. *(A,B,C,D) satisfy the following algebraic equations:*

$$\begin{pmatrix} -2\lambda d - a & 2d & 2\lambda\alpha(m,n) & -2\alpha(m,n) \\ 2\lambda^2 d & -2\lambda d - a & -2\lambda^2\alpha(m,n) & 2\lambda\alpha(m,n) \\ c & 0 & -b-2\lambda & 2 \\ 0 & c & 2\lambda^2 & -b-2\lambda \end{pmatrix} \times \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} \langle\alpha(m,n)k_{mn}(|\nabla\Phi|^2 - \lambda\Phi^2) + a\Phi^2, \Phi^2\rangle \\ \langle\alpha(m,n)k_{mn}(|\nabla\Phi|^2 - \lambda\Phi^2) + a\Phi^2, |\nabla\Phi|^2\rangle \\ 0 \\ 0 \end{pmatrix}. \tag{7.4}$$

Proof. We now recall the argument from (5.10) to (5.14). Using integration by parts, we can observe that

$$\langle\Delta\tilde{u}(0), \Phi^2\rangle = \langle\tilde{u}(0), \Delta(\Phi^2)\rangle = 2\langle\tilde{u}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle,$$
$$\langle\Delta\tilde{\rho}(0), \Phi^2\rangle = 2\langle\tilde{\rho}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle.$$

Therefore, by taking the L^2 inner product of (5.14) with Φ^2, we obtain

$$2d\langle\tilde{u}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle - a\langle\tilde{u}(0), \Phi^2\rangle - 2\alpha(m,n)\langle\tilde{\rho}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle = \langle\alpha(m,n)k_{mn}(|\nabla\Phi|^2 - \lambda\Phi^2) + a\Phi^2, \Phi^2\rangle. \tag{7.5}$$

It follows from (5.12) that

$$\begin{cases} \langle\Delta\tilde{u}(0), |\nabla\Phi|^2\rangle = -\langle\nabla\tilde{u}(0), \nabla(|\nabla\Phi|^2)\rangle \\ \quad = -2\langle\nabla\tilde{u}(0), \Delta\Phi\nabla\Phi\rangle \\ \quad = 2\langle\tilde{u}(0), \nabla(-\lambda\Phi\nabla\Phi)\rangle = 2\lambda\langle\tilde{u}(0), \lambda\Phi^2 - |\nabla\Phi|^2\rangle, \\ \langle\Delta\tilde{\rho}(0), |\nabla\Phi|^2\rangle = 2\lambda\langle\tilde{\rho}(0), \lambda\Phi^2 - |\nabla\Phi|^2\rangle. \end{cases} \tag{7.6}$$

By virtue of (7.6), taking the L^2 inner product of (5.14) with $|\nabla\Phi|^2$ leads to

$$2\lambda d\langle\tilde{u}(0), \lambda\Phi^2 - |\nabla\Phi|^2\rangle - a\langle\tilde{u}(0), |\nabla\Phi|^2\rangle - 2\lambda\alpha(m,n)\langle\tilde{\rho}(0), \lambda\Phi^2 - |\nabla\Phi|^2\rangle = \langle\alpha(m,n)k_{mn}(|\nabla\Phi|^2 - \lambda\Phi^2) + a\Phi^2, |\nabla\Phi|^2\rangle. \tag{7.7}$$

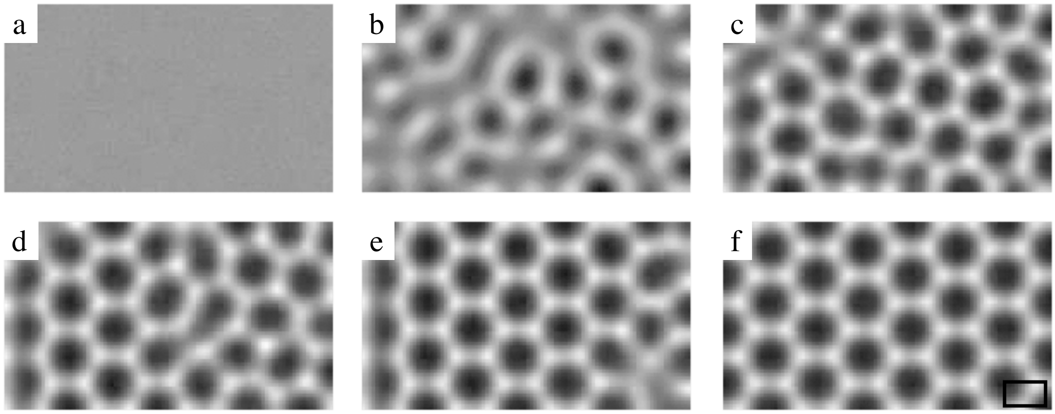

Fig. 1. Temporal development of the variable u with gray levels, where black and white indicate $u = 0.0$ and 2.0, respectively. (a) $t = 0$, (b) $t = 15.259$, (c) $t = 30.518$, (d) $t = 152.588$, (e) $t = 305.176$, (f) $t = 457.764$. The rectangle in (f) shows the unit cell size having the packing number, 7, for hexagons. The initial values (u_0, v_0) are $(1 + n(x, y), 1/32)$. The perturbation $n(x, y)$ is white noise between -0.05 and 0.05. The numerical integration scheme used is explicit in time with central difference approximations for the diffusion terms and the advection term. The spatial grid sizes are $\Delta x = \Delta y = 2^{-5}$ and the time step is $\Delta t = 2^{-16}$.

Differentiating the second equation of (SP) twice with respect to s and setting $s = 0$, we have

$$\Delta\tilde{\rho}(0) - b\tilde{\rho}(0) + c\tilde{u}(0) = 0. \tag{7.8}$$

From (5.14) and (7.6), the L^2 inner products of (7.8) with Φ^2 and $|\nabla\Phi|^2$ enable us to obtain

$$2\langle\tilde{\rho}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle - b\langle\tilde{\rho}(0), \Phi^2\rangle + c\langle\tilde{u}(0), \Phi^2\rangle = 0 \tag{7.9}$$

and

$$2\lambda\langle\tilde{\rho}(0), \lambda\Phi^2 - |\nabla\Phi|^2\rangle - b\langle\tilde{\rho}(0), |\nabla\Phi|^2\rangle + c\langle\tilde{u}(0), |\nabla\Phi|^2\rangle = 0, \tag{7.10}$$

respectively. From (7.5), (7.7), (7.9) and (7.10), (A, B, C, D) defined by (7.2) satisfies (7.4). The proof of Lemma 7.2 is complete. □

Proof of Theorem 7.1. By differentiating (5.13) with respect to s, we obtain

$$\begin{aligned} &d\Delta u'''(s) - \alpha'''(s)\nabla\{u(s)\nabla\rho(s)\} \\ &\quad - 3\alpha''(s)\nabla\{u'(s)\nabla\rho(s) + u(s)\nabla\rho'(s)\} \\ &\quad - 3\alpha'(s)\nabla\{u''(s)\nabla\rho(s) + 2u'(s)\nabla\rho'(s) + u(s)\nabla\rho''(s)\} \\ &\quad - \alpha(s)\nabla\{u'''(s)\nabla\rho(s) \\ &\quad + 3u''(s)\nabla\rho'(s) + 3u'(s)\nabla\rho''(s) + u(s)\nabla\rho'''(s)\} \\ &\quad + au'''(s)\{1 - 2u(s)\} - 6au'(s)u''(s) = 0. \end{aligned} \tag{7.11}$$

As $s = 0$ in (7.11), it follows that

$$\begin{aligned} &d\Delta\tilde{u}'(0) - k_{mn}\tilde{\alpha}(0)\Delta\Phi - \alpha(m, n)\nabla\{k_{mn}\tilde{u}(0)\nabla\Phi \\ &\quad + \Phi\nabla\tilde{\rho}(0) + \nabla\tilde{\rho}'(0)\} - a\tilde{u}'(0) - 2a\tilde{u}(0)\Phi = 0. \end{aligned} \tag{7.12}$$

By taking the L^2 inner product of (7.12) with Φ, we have

$$\begin{aligned} -k_{mn}\tilde{\alpha}(0)\langle\Delta\Phi, \Phi\rangle &= -d\langle\Delta\tilde{u}'(0), \Phi\rangle + \alpha(m, n) \\ &\quad \times \langle\nabla\{k_{mn}\tilde{u}(0)\nabla\Phi + \Phi\nabla\tilde{\rho}(0) + \nabla\tilde{\rho}'(0)\}, \Phi\rangle \\ &\quad + a\langle\tilde{u}'(0), \Phi\rangle + 2a\langle\tilde{u}(0), \Phi^2\rangle. \end{aligned}$$

It follows from (5.12) and $\langle\Delta\tilde{u}'(0), \Phi\rangle = -\lambda\langle\tilde{u}'(0), \Phi\rangle = 0$ that

$$\begin{aligned} \lambda k_{mn}\tilde{\alpha}(0)\|\Phi\|^2 &= \alpha(m, n)\langle\nabla\{k_{mn}\tilde{u}(0)\nabla\Phi + \Phi\nabla\tilde{\rho}(0) \\ &\quad + \nabla\tilde{\rho}'(0)\}, \Phi\rangle + 2a\langle\tilde{u}(0), \Phi^2\rangle. \end{aligned} \tag{7.13}$$

Noting

$$\begin{aligned} &\langle\nabla\{\tilde{u}(0)\nabla\Phi\}, \Phi\rangle = -\langle\tilde{u}(0)\nabla\Phi, \nabla\Phi\rangle = -\langle\tilde{u}(0), |\nabla\Phi|^2\rangle, \\ &\langle\nabla\{\Phi\nabla\tilde{\rho}(0)\}, \Phi\rangle = -\langle\Phi\nabla\tilde{\rho}(0), \nabla\Phi\rangle = -\langle\nabla\tilde{\rho}(0), \Phi\nabla\Phi\rangle \\ &\quad = \langle\tilde{\rho}(0), \nabla\{\Phi\nabla\Phi\}\rangle = \langle\tilde{\rho}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle, \\ &\langle\Delta\tilde{\rho}'(0), \Phi\rangle = \langle\tilde{\rho}'(0), \Delta\Phi\rangle = -\lambda\langle\tilde{\rho}'(0), \Phi\rangle = 0, \end{aligned}$$

we have

$$\begin{aligned} \lambda k_{mn}\tilde{\alpha}(0)\|\Phi\|^2 &= \alpha(m, n)\{\langle\tilde{\rho}(0), |\nabla\Phi|^2 - \lambda\Phi^2\rangle \\ &\quad - k_{mn}\langle\tilde{u}(0), |\nabla\Phi|^2\rangle\} + 2a\langle\tilde{u}(0), \Phi^2\rangle. \end{aligned} \tag{7.14}$$

From (7.2) and (7.14), we obtain (7.3). Then Theorem 7.1 is completely proved. □

8. Numerical results

In this section, we exhibit various numerical results for (P). To do so, let (a, α) be parameters and fix other constants

$$d = 0.0625\left(= \frac{1}{16}\right), \qquad b = 32, \qquad c = 1. \tag{8.1}$$

The initial values (u_0, v_0) are taken by positive constant solution $(1, c/b)$ with some perturbations and the horizontal length of each rectangle is 6, that is,

$$\Omega = (0, 6) \times \left(0, \frac{6}{\sqrt{3}}\right).$$

Fig. 1 shows a typical time-dependent behavior of a numerical solution for $u(x, t)$ which tends to a hexagonal steady state as t grows. Fig. 2 shows snap shots of numerical solutions for $u(x, t)$ with a sufficiently large t in the parameter space (a, α).

When $(a, \alpha) = (7, 17)$, the packing number of hexagons in Ω is 7 in the horizontal direction (x-axis) in Fig. 1. The profile of this hexagonal pattern is similar to the function

$$\Phi = -\cos(2\pi x) - 2\cos\pi x\cos(\sqrt{3}\pi y). \tag{8.2}$$

Therefore, the unit cell size of the hexagonal pattern is represented by

$$\left(0, \frac{6}{7}\right) \times \left(0, \frac{6}{7\sqrt{3}}\right).$$

Restricted to the unit rectangle, we know that the function Φ in (8.2) coincides with that in (6.11) for

$$(m, n) = (2, 0),\ (1, 1) \quad \text{and} \quad l = \frac{7\pi}{6}. \tag{8.3}$$

Hence, (4.8) implies that the branch of hexagonal solutions bifurcates from the constant solution $(u, \rho) = (1, c/b)$ when (a, α) satisfies

$$\alpha = \frac{\{dl^2(m^2 + 3n^2) + a\}\{l^2(m^2 + 3n^2) + b\}}{cl^2(m^2 + 3n^2)}\ (=: \alpha(m, n)) \tag{8.4}$$

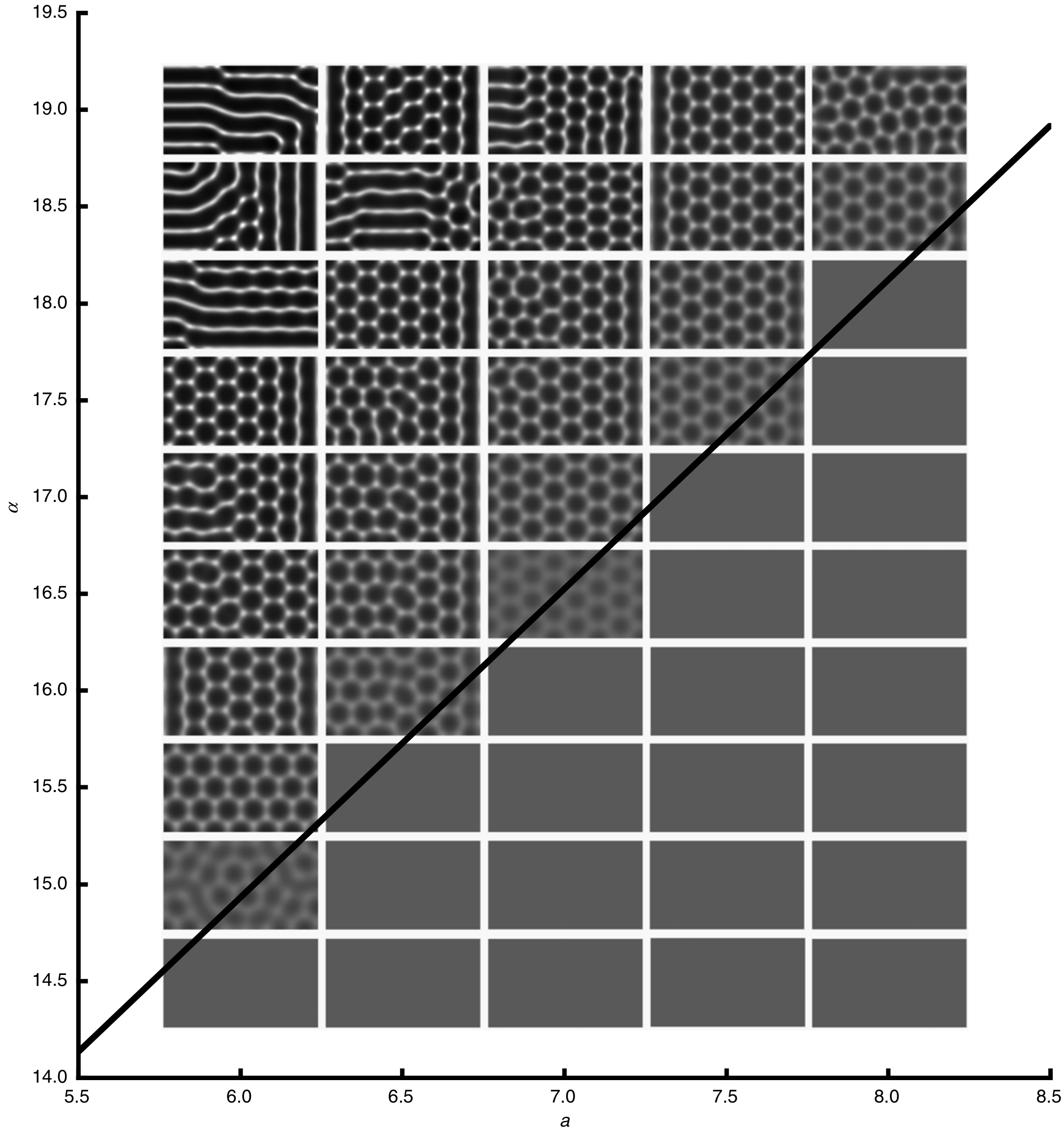

Fig. 2. Patterns of u at $t = 1000$ in the parameter space (a, α). The line corresponds to the bifurcation line of the hexagon given by (8.5).

with (8.1) and (8.3). By substituting (8.1) and (8.3) into (8.4), we obtain

$$\alpha = \left(1 + \frac{288}{49\pi^2}\right)\left(a + \frac{49\pi^2}{144}\right), \tag{8.5}$$

which is the line shown in Fig. 2. As shown, the line is in the neighborhood of the numerical bifurcation line in the parameter space (a, α). On the other hand, the direction of the bifurcation branch corresponding to the hexagons is given by $\beta(0)$ in (6.10). Since $\beta(0) = (l^2 + 8)\,(4a - l^2)/8l^2$ for $l = 7\pi/6$, we have that if $a > 49\pi^2/144$, then $\beta(0) > 0$. Therefore, if $a > 49\pi^2/144$, then the bifurcation branch corresponding to hexagonal patterns for small $s < 0$ exists in the region below the line in Fig. 2. Although this pattern is unstable in the neighborhood of the bifurcation point, there exist stable hexagonal patterns in the upper region. We conjecture that this unstable branch turns at some point $\alpha^* < \alpha(1, 1)$ for fixed a and recovers its stability (see [28,21]). Since the initial value for the numerical computations is restricted, we cannot derive the global bifurcation diagram of the stationary solutions from Fig. 2.

9. Concluding remarks

The main aim of the present paper is to analyze the existence of solutions of (SP), as well as their dependency on the intensity of the advection and growth. In Section 3, we show the asymptotic behavior of positive solutions of (SP) as $\alpha \to \infty$. For the result of Theorem 3.2, it is worth noting that some numerical results by Kurata et al. [17] support the existence of $\{(u_n, \rho_n)\}$ satisfying (i) of the theorem. In particular, some bifurcation diagrams performed by AUTO (Figs. 8–10 in [17]) imply the branches of stationary solutions (u_α, ρ_α) such that $\|u_\alpha\|_2 + \|\rho_\alpha\|_2$ decays along the branches when α grows. Results from numerical computation in [17,20] suggest the existence of the spike solution of (SP), which corresponds to the solution of case (i). Then, the future work will be to investigate the existence of sequence $\{(u_n, \rho_n)\}$ satisfying (i), and the asymptotic behavior of the location and height of spikes of u_n as $n \to \infty$, namely, $\{x \in \overline{\Omega} : u_n(x) = \max_{\overline{\Omega}} u_n\}$ and $\max_{\overline{\Omega}} u_n$. We refer the reader to [27] for the construction of peak solutions to a related problem.

In Section 8, we clarify the dynamics of pattern formations by numerical computation. The dynamics of two-dimensional

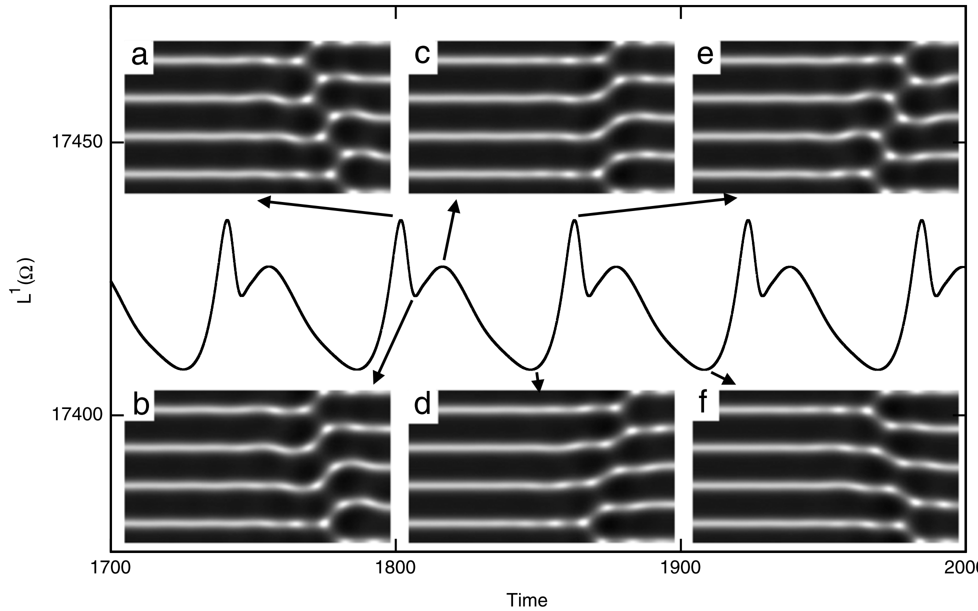

Fig. 3. Unstable defective stripe pattern and temporal evolution of the L^1-norm of u with $a = 5.0$ and $\alpha = 16.5$ [29]. (a) $t = 1801.758$, (b) $t = 1806.793$, (c) $t = 1816.406$, (d) $t = 1847.382$, (e) $t = 1862.640$, (f) $t = 1908.264$.

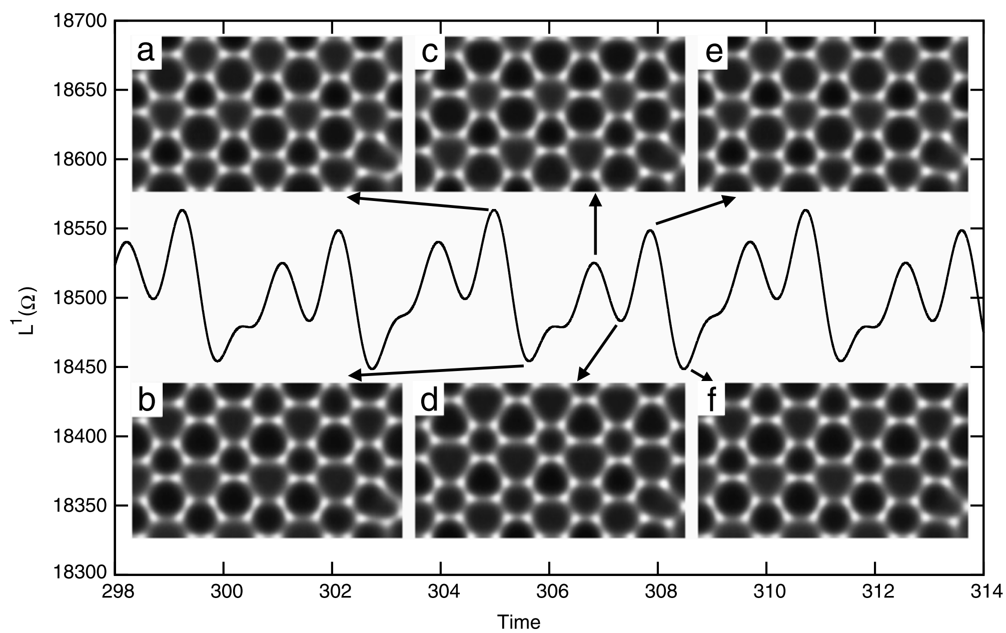

Fig. 4. Rapid breathing-like hexagonal pattern and temporal evolution of the L^1-norm of u with $a = 8.0$ and $\alpha = 21.0$ [29]. (a) $t = 305.023$, (b) $t = 305.634$, (c) $t = 306.854$, (d) $t = 307.312$, (e) $t = 307.922$, (f) $t = 308.533$.

patterns fall into multiple classes according to their spatio-temporal properties. The first class is the homogeneous pattern, in which the initial values with the perturbation quickly set to the unique positive constant solution $(u, \rho) = (1, c/b)$, under the bifurcation line for the hexagon. The second is the stable hexagonal pattern, which appeared near the bifurcation line. The third is the spatial irregular pattern having hexagons and stripes, which appears for parameter values far above the bifurcation line.

Finally, we discuss spatio-temporal patterns. The appearance of oscillating spike patterns in the one-dimensional interval has previously been shown in [17,20]. Fig. 3 implies that there is a spatio-temporal pattern in which the connections between stripes periodically change, that is, unstable defective stripe patterns, where the stripe patterns are stable. Moreover, in Fig. 4 we show the rapid breathing oscillation of the hexagons and will give some movies of the numerical results in the supplementary content [29]. It is important to study the spatio-temporal periodic pattens in view of instability of the stationary solutions.

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Appendix. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.physd.2012.06.009.

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[29] Some movies of the numerical results are available in the supplementary content.